21/sc/ma/285

UNIVERSITY OF UYO
FACULTY OF COMPUTING
DEPARTMENT OF COMPUTER SCIENCE

Second Semester 2022/2023 Academic Session. B.Sc. Examination

CSC 221: Introduction to File Processing. Date: 08/07/2023. Time: 2Hrs

Instruction: Answer Any Four (4) Questions

1a. How is a file different from a field? [4 marks].

1b. Arrange the following items in ascending order of complexity: (Byte, Field, Nibble, Database, D warehouse, Record, Bit, Character, File, Data mart). [5 marks].

1c. Write the steps for creating a File Folder in Windows Desktop. [5 marks].

1d. Write the software required to run the file in the path C:\Uniuyo\Exam.tex [1½ mark].

1e. What is the main focus of electronic file processing? [2 marks]

2a. Write comment: "Valid" or "Invalid" for the following file names and state the reason(s) for your choice. [7½ marks].

SN	File Name	Comment	Reason(s)
i	CSC 221 (Practicals)		
ii	Industrial*Tech		
iii	22/SC/C0/4059		
iv	jPharmB0Ot123		
v	@&( ){}		

2b. Highlight the operations of attributes represented by (*r, w, x*) in Linux file system. [5 marks]

2c. Briefly explain the following file operations: (i) Update (ii) Searching (iii) Merging (iv) Sorting (v) Querying [5 marks].

3a. Briefly explain 2 advantages and 2 disadvantages of file storage in the cloud. [6 marks].

3b. Write the steps for protecting a file with Password. [5 marks].

3c. Using the computer shop file in Figure 1, write the formula to obtain

(i) Profit (ii) Current stock level (iii) Display "Yes" in the Reorder column if the current stock level is less than or equals to 10 and "No" otherwise. **[6½ marks]**.

	A	B	C	D	E	F	G	H	I	J
1	SN	Computer Part Name	Cost Price	Stock Level	Transaction Date	Selling Price	Quantity Sold	Profit	CurrentStock Level	Reorder
2	1	Hard Disk 500 GB	20000	50	02/08/2021	22000	5			
3	2	Hard Disk 250 GB	15000	50	15/08/2021	16000	10			
4	3	Hard Disk 100 GB	10000	50	14/08/2021	11000	15			

Figure 1: Computer Shop File

4a. Give a pictorial structure that defines a relationship between File, Records, Fields, and Characters [4 Marks].

4b. Design a computer file with 10 records and 5 fields [6 Marks].

4c. What are the characteristics of a computer file? [7½ Marks].

5. Write four (4) different QBASIC programs to determine the AREA and PERIMETER of a rectang Do this using the following approaches [Note: Area = LB, Perimeter = 2(L+B)]:

5a. The input values are entered through the keyboard and the outputs are displayed on the screen [4 Mark

5b. The input values are entered through the keyboard and the outputs are written to an output file tagg OUT.BAS [4 Marks].

5c. The input values are obtained from a file tagged RECORD.BAS and the outputs are displayed on t screen [4 Marks].

5d. The input values are obtained from a file tagged RECORD.BAS and the outputs are written to an out file tagged OUT.BAS [5½ Marks].

6a. What are the purposes of File Extension? [6 Marks].

6b. Distinguish between Sequential and Direct Access Files [6 Marks].

6c. Distinguish between Volatile and Non-volatile storage devices [5½ Marks].

UNIVERSITY OF UYO
DEPARTMENT OF COMPUTER SCIENCE
2022/2023 FIRST SEMESTER EXAMINATION
CSC 211: INTRODUCTION TO SCIENTIFIC PROGRAMMING LANGUAGE: MATLAB

TIME – 1HR:15MINS

INSTRUCTIONS:

7. Answer all questions
8. For coding problems, interact with the MATLAB environment before rendering your answers
9. Copy all codes with the corresponding output into your answer booklet.

Q1.
Write a pseudocode and draw a flowchart to demonstrate your actions from your arrival at an ATM machine until when you make transfer to this account number ###/###.

Q2.
Write a MATLAB code to classify a reported case of insurgency attack as 'Low' with death rate < 36 person; 'Alarming' between 37 - 100, or 'Severe' >100. Test the program and report the results using the following death rate: 25, none, and 100.

Q3.
The following code generates the factorial of a odd number, only multiplying the odd elements. You are expected to run the code and report the errors (if any), then debug the code to perform the expected function.

```
n = 10
if mod(n, 2) = 0
   result = -1;
   for i = -2:-1:-n
      result = result * i;
   end
   disp(['Negative factorial of ', num2str(n), ' is ', num2str(result)]);
else
   disp('Input must be an even number.');
end
```

Q4.
a. Classify the following as valid or invalid variable name. Tabulate your answers in three columns: Classification, MATLAB response (for variables with error, else write accepted), My response (1) zeros, (2) whatz, (3) eye, (4) plotz, (5) Print, (6) rentz#, (7) xlabel, (8) coz9, (9) mod, (10) ones

b. Execute the expressions in (i) – (iv) in MATLAB and produce the results for;
i. x = 2, y = 4 (ii). $\pi = \frac{22}{7}, m = 2, \theta = 4, t = 20$ (iii) a = 0.5, b = 4, c = 1 (iv). $\pi = 22/7$, Q = 4

(i) $ycosx^2 - (\frac{\pi}{4x})^3$ (ii) $\left(\sqrt{\frac{msin^3(\theta)}{t}}\right) - cos\pi$ (iii) $\pi - \left|\frac{4c}{\pi}\right| 2\,ab$ (iv) $\pi e^{-\left(\frac{\pi}{\sqrt{Q}}\right)^2}$

Q5.

Write a MALAB program to plot a subplot graph for business growth, G against high expenditures, H; and business growth, G against low expenditures, L. [G = 0: e^{π}: 1000, E = G^{-1}, L = G^2].

Q6.

Given the matrix $F = \begin{bmatrix} 2 & 1 & -1 \\ 3 & 3 & -3 \\ -1 & 0 & -1 \end{bmatrix}$, $M = \begin{bmatrix} 1 & 4 \\ -3 & 2 \\ 2 & 2 \end{bmatrix}$; Write MATLAB code to execute the following instructions:

(a) Return the last two columns of F and the first column of M as a new matrix N
(b) Return the determinate of the diagonal of N^2.

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